TIME SERIES ANALYSIS

DATA MANAGEMENT AND ANALYSIS

EMPIRICL PAPER :

INTEREST RATE TRENDS AND PRIVATE SECTOR CREDIT GROWTH IN GHANA: AN EMPIRICAL TIME SERIES ASSESSMENT (2003 - 2022)

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Table of Contents

List of Tables 3

List of Figures 4

Abstract 5

1. Introduction 6

1.1 Objectives 6

1.2 Brief Literature Review 7

2. Theoretical Framework and Methodology 8

2.1 Theoretical expectation and model 8

2.2 Data and estimation 8

3. Descriptive Statistics and Trends 9

4. Empirical Results and Discussion 12

4.1 Unit-root tests 12

4.2 ARDL bounds test and long-run association 12

4.3 Diagnostic and stability tests 14

4.4 VAR and Granger causality 16

5. Conclusion and Policy Implications 17

5.1 Limitations and Future Research 17

References 18

Appendix 20

Diagnostic figure 20

List of Tables

Table 1: Descriptive Statistics of Monthly Series 9

Table 2: ADF and Phillips–Perron unit-root results 12

Table 3: Table 3: ARDL Model and Bounds Test 13

Table 4: Breusch-Godfrey LM Test for Serial Correlation 15

Table 5: ARCH LM Test for Conditional Heteroskedasticity 15

Table 6: Pairwise Granger Causality Wald Test 16

List of Figures

Figure 1: Nominal private sector credit in Ghana, 2003–2022 10

Figure 2: Logarithm of private sector credit, 2003–2022 10

Figure 3: Average commercial bank lending rate, 2003–2022 11

Figure 4: Logged private sector credit and lending rate, 2003–2022 11

Abstract

This study examines the relationship between commercial bank lending rates and private sector credit in Ghana using monthly data from January 2003 to December 2022 (240 observations). Augmented Dickey–Fuller and Phillips–Perron tests indicate that logged private sector credit, and the lending rate are integrated of order one I(1). An ARDL(2,1) bounds test produces an F-statistic of 10.882, above the 1% upper-bound critical value of 7.84, supporting a long-run relationship. The estimated long-run coefficient of −0.1894 implies that a one-percentage-point increase in the lending rate is associated with approximately 17.25% lower nominal private sector credit in the long run. The error-correction coefficient of −0.0072 indicates slow monthly adjustment of about 0.72%. Granger tests find no statistically significant short-run predictive relationship in either direction. Heteroskedasticity and a significant Ramsey RESET test indicate that model limitations remain and inference should be cautious. The findings describe an association rather than a causal effect.

Keywords: commercial bank lending rate; private sector credit; ARDL; error correction; Granger causality; Ghana.

1. Introduction

Private sector credit supports business expansion, capital formation and employment, while high borrowing costs can restrict firms' access to finance (Levine, 2005; Beck et al., 2007). In Ghana, commercial banks are an important source of private financing, yet lending rates have remained relatively high and credit conditions are influenced by macroeconomic and banking - sector developments. Between 2003 and 2022, Ghana experienced changes in monetary policy, a banking - sector clean-up, the COVID-19 shock and a severe macroeconomic crisis in 2022 (Bank of Ghana, 2022). Over the same period, nominal private sector credit increased substantially, while commercial bank lending rates fluctuated considerably.

The credit channel of monetary transmission suggests that changes in monetary and funding conditions can influence commercial lending rates, credit demand and banks' willingness to lend (Bernanke & Blinder, 1988; Bernanke & Gertler, 1995). Higher rates raise the cost of capital and may reduce borrowing, although banks can also ration credit when information asymmetry and default risk are high (Stiglitz & Weiss, 1981). The direction and strength of these relationships are empirical questions.

1.1 Objectives

The study has four objectives:

1. Assess the stationarity of private sector credit and lending rates.

2. Test for a long-run relationship between the variables.

3. Estimate the long-run association and speed of adjustment, and

4. Test whether changes in lending rates and credit growth predict one another in the short run.

1.2 Brief Literature Review

Neoclassical investment theory links higher interest rates to a higher user cost of capital and weaker investment demand (Jorgenson, 1967). The credit-channel literature also argues that monetary policy affects bank lending and aggregate demand (Bernanke & Blinder, 1988; Bernanke & Gertler, 1995). Stiglitz and Weiss (1981) show that imperfect information can lead banks to ration credit instead of clearing the market through interest rates alone. Research on emerging economies identifies domestic lending rates and monetary conditions as important correlates of credit growth (Guo & Stepanyan, 2011). Ghana-focused research has also examined lending-rate spreads and monetary policy transmission (Amidu, 2006; Bawumia et al., 2008). This study contributes a monthly analysis over 2003–2022 using unit-root tests, ARDL bounds testing and Granger causality analysis.

2. Theoretical Framework and Methodology

2.1 Theoretical expectation and model

The study expects an inverse relationship between commercial bank lending rates and private sector credit. Banks' lending rates reflect funding costs, operating costs, risk premia and mark-ups as well as broader monetary conditions. The baseline long-run relationship is:

$\log (PSC_t) = \beta_0 + \beta_1 \, LENDING_RATE_t + \varepsilon$, with $\beta_1 < 0$

PSC is nominal private sector credit in millions of Ghana cedis; log (PSC) is its natural logarithm; and LENDING_RATE is the average commercial bank lending rate measured in percentage points. Because the dependent variable is logged but the lending rate is not, β_1 is a semi-elasticity, not an elasticity. The model is bivariate and does not isolate the effect of the Bank of Ghana policy rate or control for other macroeconomic factors.

2.2 Data and estimation

Monthly data from January 2003 to December 2022 were obtained from the Bank of Ghana's monetary and financial statistics (Website). The dataset contains 240 observations for nominal private sector credit and the average commercial bank lending rate. The analysis uses three techniques:

1. ADF and PP unit-root tests at levels and first differences;
2. An ARDL bounds test and error-correction representation; and
3. A VAR in first differences with four monthly lags, followed by Granger causality tests. The ARDL approach is suitable when variables are I(0) or I(1), but not I(2) (Pesaran et al., 2001).

3. Descriptive Statistics and Trends

Table 1 summarises the main variables. Nominal credit has a much larger standard deviation than logged credit, consistent with a strong upward trend in the level series. The commercial bank lending rate averages 26.85%, with a minimum of 20.04% and a maximum of 36.50%. Monthly credit growth is volatile and negatively skewed, with excess kurtosis, indicating occasional large movements. The normality tests reject normality for each series; these are tests of the variables' distributions, not the model residuals, and do not alone invalidate ARDL estimation.

Table 1: Descriptive Statistics of Monthly Series

	Mean	Median	SD	Min	Max	Skewness	Kurtosis	N
psc	18500	11500	1.72e+04	552.25	73700	.8038	2.7813	240
log psc	9.1357	9.3544	1.3827	6.314	11.2084	-.4796	1.956	240
lending rate	26.8463	26.72	3.6108	20.04	36.5	.4753	3.0949	240
d log psc	.0196	.0204	0.0285	-.1456	.1292	-1.3559	10.0625	239

Source: Estimations using Stata.

Figures 1–3 show that nominal and logged private sector credit generally increased over the sample, while lending rates moved in cycles and rose sharply near the end of 2022. The level-series increase should not be interpreted as an equivalent increase in real credit because the measure is nominal. The sharp decline in the final monthly credit observation also merits verification against the original Bank of Ghana series before it is interpreted as an actual contraction in credit availability.

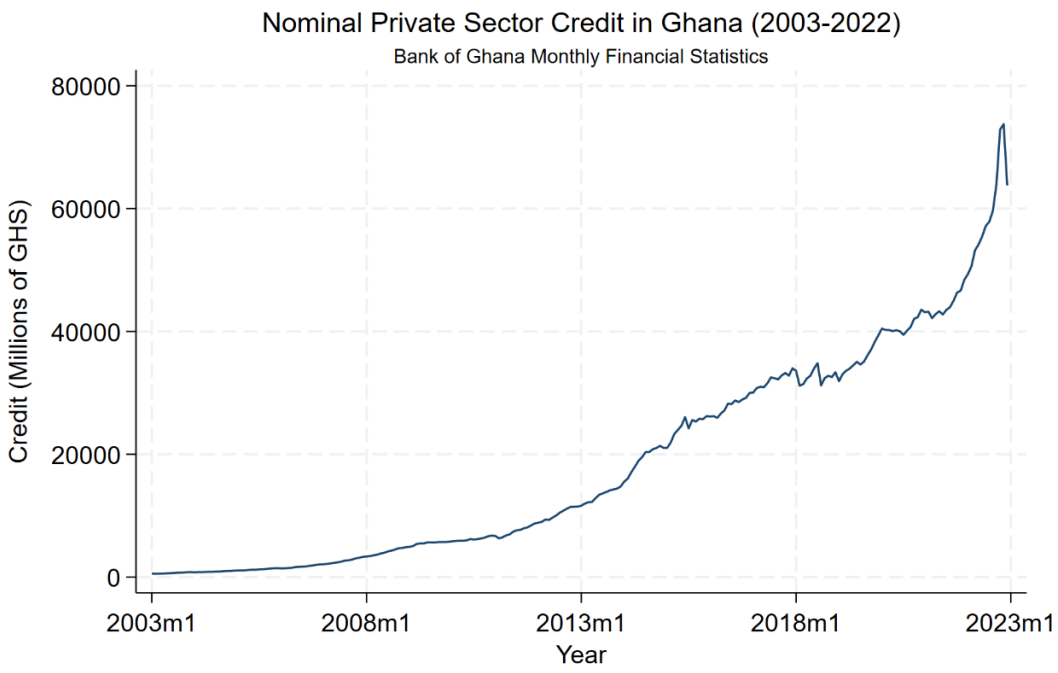


Figure 1: Nominal private sector credit in Ghana, 2003–2022

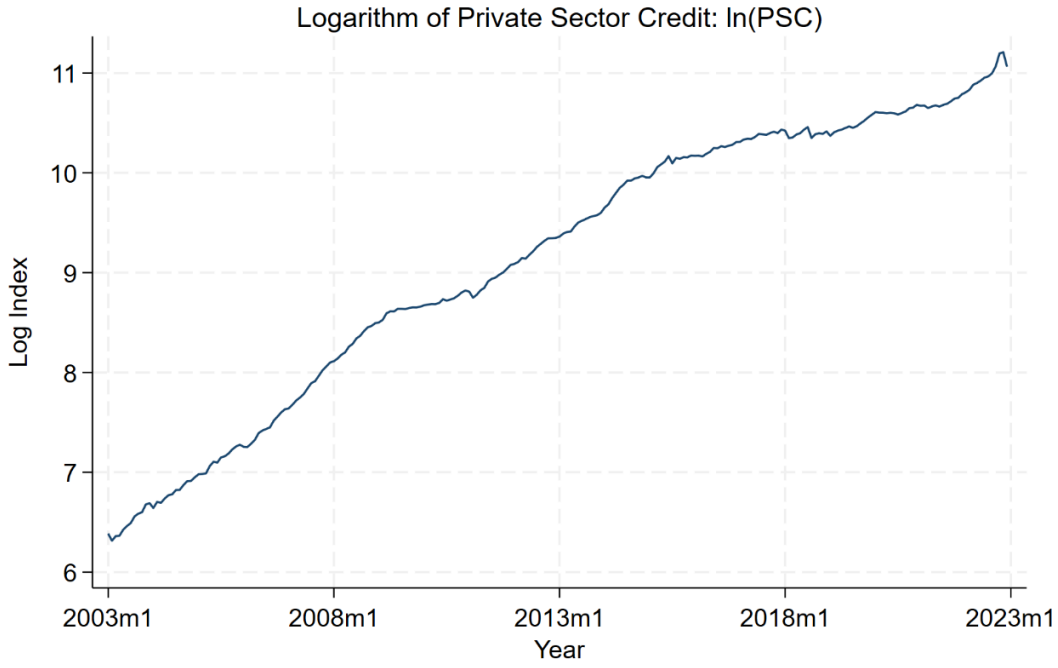


Figure 2: Logarithm of private sector credit, 2003–2022

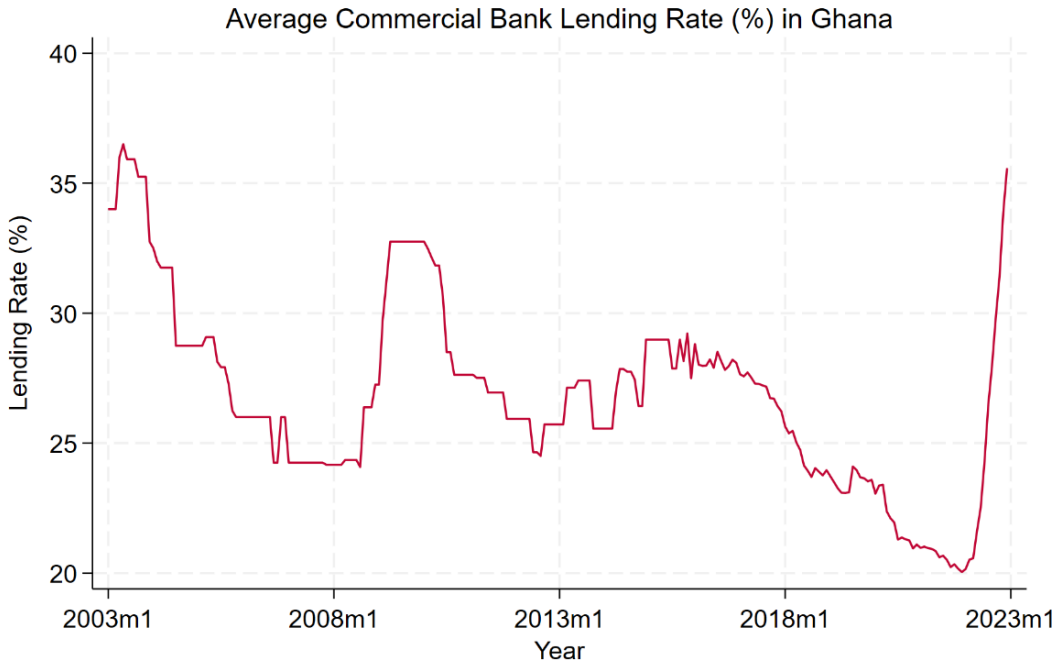


Figure 3: Average commercial bank lending rate, 2003–2022

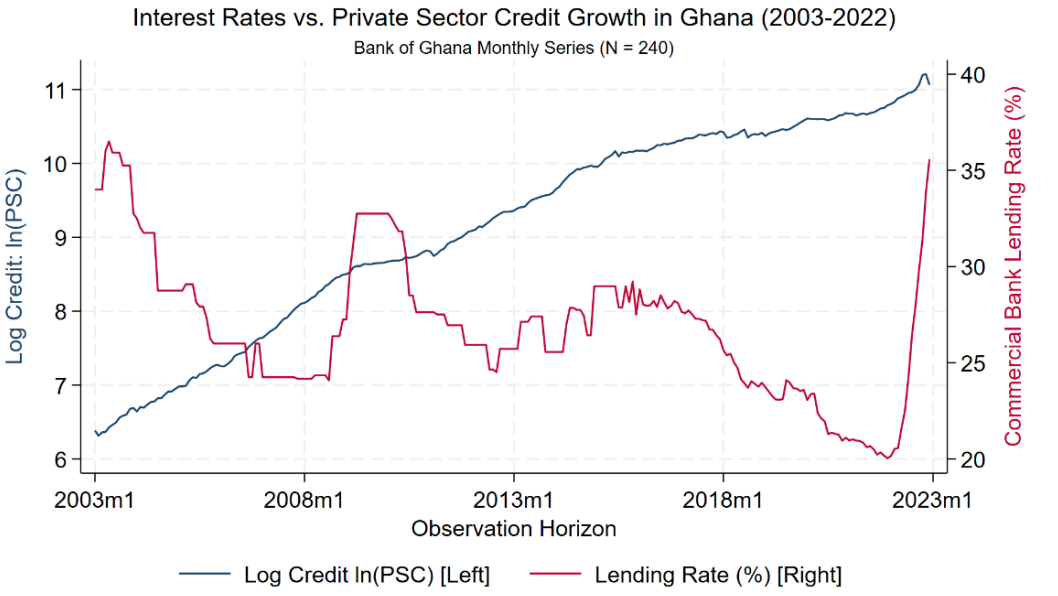


Figure 4: Logged private sector credit and lending rate, 2003–2022

4. Empirical Results and Discussion

4.1 Unit-root tests

At levels, both log (PSC) and the lending rate fail to reject the unit-root null under the reported Augmented Dickey-Fuller (ADF) and Philips-Perron (PP) tests. Their first differences are stationary at the 1% level. The results therefore classify both variables as I(1), satisfying the integration-order requirement for ARDL bounds testing, provided no variable is I(2).

Table 2: ADF and Phillips–Perron unit-root results

Variable	Specification	ADF statistic (p)	PP statistic (p)	Conclusion
log(PSC), level	Constant & trend	-1.133 (0.9233)	-0.830 (0.9631)	Non-stationary
Δlog(PSC)	Constant	-9.452 (< 0.000) ***	-13.524 (< 0.000) ***	Stationary I(1)
Lending rate, level	Constant	-1.953 (0.3073)	-1.869 (0.3470)	Non-stationary
Δ lending rate	Constant	-7.425 (< 0.001) ***	-12.143 (< 0.001) ***	Stationary I(1)

4.2 ARDL bounds test and long-run association

The selected ARDL (2,1) model uses 238 observations. Its bounds-test F-statistic is 10.882, above the 1% upper-bound critical value of 7.84 for Case 3 with one long-run regressor. The null hypothesis of no levels relationship is therefore rejected at the 1% level. The reported t-statistic of -4.664 is also below the 1% I(1) critical value of -3.820, providing additional support for a levels relationship.

ARDL (2,1) regression
Sample: 2003m3 thru 2022m12

Number of obs = 238
R-squared = 0.0919
Adj R-squared = 0.0763
Root MSE = 0.0269

Log likelihood = 525.38138

Table 3: Table 3: ARDL Model and Bounds Test

D.log_psc	Coefficient	Std.	err.	t	P>t	[95%	conf.	interval]
ADJ								
log_psc								
L1.	-0.007	0.002	-4.660	0.000	-0.010	-0.004		
LR								
lending_rate	-0.189	0.069	-2.760	0.006	-0.325	-0.054		
SR								
log_psc								
LD.	0.015	0.068	0.210	0.831	-0.119	0.148		
lending_rate								
D1.	0.005	0.003	1.790	0.074	-0.000	0.010		
_cons	0.122	0.026	4.670	0.000	0.070	0.173		

note: estat btest has been superseded by estat ectest
as the prime procedure to test for a levels relationship.
(click to run)

Pesaran/Shin/Smith (2001) ARDL Bounds Test
H0: no levels relationship F = 10.882
t = -4.664
Critical Values (0.1-0.01), F-statistic, Case 3

[I_0]	[I_1]	[I_0]	[I_1]	[I_0]	[I_1]	[I_0]	[I_1]
	L_1	L_1	L_05	L_05	L_025	L_025	L_01	L_01
k_1 4.040	4.780	4.940	5.730	5.770	6.680	6.840	7.840	

accept if F < critical value for I(0) regressors
reject if F > critical value for I(1) regressors
Critical Values (0.1-0.01), t-statistic, Case 3

[I_0]	[I_1]	[I_0]	[I_1]	[I_0]	[I_1]	[I_0]	[I_1]
	L_1	L_1	L_05	L_05	L_025	L_025	L_01	L_01
k_1	-2.570	-2.910	-2.860	-3.220	-3.130	-3.500	-3.430	-3.820

accept if t > critical value for I(0) regressors
reject if t < critical value for I(1) regressors
k: of non-deterministic regressors in long-run relationship
Critical values from Pesaran/Shin/Smith (2001)

The long-run lending-rate coefficient is negative and statistically significant. Since the dependent variable is log (PSC) and the lending rate is in percentage points, a one-percentage-point increase in the lending rate is associated with an estimated $100 \times [\exp(-0.1894) - 1] \approx 17.25\%$ lower long-run nominal private sector credit. This is an association conditional on the bivariate specification, not proof that a lending-rate increase causes credit to fall. The short-run lending-rate change is not significant at 5% ($p = .074$), so immediate effects are less conclusive.

The adjustment coefficient is negative and statistically significant. Its magnitude implies that about 0.72% of the preceding period's disequilibrium is corrected each month, indicating slow adjustment towards the estimated long-run path. This estimate should be interpreted cautiously because diagnostic tests identify heteroskedasticity and possible functional-form misspecification.

4.3 Diagnostic and stability tests

Breusch - Godfrey tests do not reject the null of no serial correlation at lags 1, 2 or 4 ($p = 0.747$, 0.393 and 0.267). However, the ARCH LM test rejects no ARCH effects at lag 2 ($\chi^2 = 30.477$, $p < 0.001$), and the Breusch–Pagan/Cook–Weisberg test rejects constant variance ($\chi^2 = 29.29$, $p < 0.000$). The Ramsey RESET test is significant ($F(3,230) = 5.87$, $p = 0.0007$), indicating possible functional-form misspecification or omitted nonlinear terms. The reported CUSUM statistic (0.6921) is below the 10%, 5% and 1% critical values, so this test does not indicate parameter instability. These diagnostics do not support treating all conventional standard errors as fully reliable; robust inference and model re-specification was also considered.

Breusch–Godfrey LM Test for Serial Correlation

Table 4: Breusch-Godfrey LM Test for Serial Correlation

Breuschâ€ “Godfrey LM test for autocorrelation chi2	df	Prob>Chi2
0.104	1	0.747
1.866	2	0.393
5.205	4	0.267

H0: no serial correlation

Autoregressive Conditional Heteroskedasticity Test (ARCH LM)

Table 5: ARCH LM Test for Conditional Heteroskedasticity

LM test for autoregressive conditional heteroskedasticity (ARCH) chi2	df	Prob>Chi2
0.403	1	0.525
30.477	2	0.000

H0: no ARCH effects vs. H1: ARCH(p) disturbance

Cumulative sum test for parameter stability

Sample: 2003m3 thru 2022m12 Number of obs = 238

H0: No structural break

	Test	-------- Critical value ---------		
Type	statistic	1%	5%	10%
Recursive	0.6921	1.1430	0.9479	0.8499

4.4 VAR and Granger causality

A VAR in first differences with four monthly lags was estimated using 235 observations. The joint Granger tests find no statistically significant predictive relationship in either direction: changes in lending rates do not Granger-predict credit growth ($\chi^2 = 2.242$, $p = 0.691$), and credit growth does not Granger-predict changes in lending rates ($\chi^2 = 2.873$, $p = 0.579$). Thus, the null of no Granger causality cannot be rejected in either direction. This is not proof that the variables are unrelated; it indicates that four lags of one differenced series do not add statistically significant predictive information for the other in this specification.

Table 6: Pairwise Granger Causality Wald Test

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob>Chi2
d_log_psc	d_lending_rate	2.242	4	0.691
d_log_psc	ALL	2.242	4	0.691
d_lending_rate	d_log_psc	2.873	4	0.579
d_lending_rate	ALL	2.873	4	0.579

5. Conclusion and Policy Implications

Using monthly Ghanaian data for 2003 - 2022, this study finds that logged private sector credit and commercial bank lending rates are I(1), and the ARDL bounds test supports a long-run relationship between them. The estimated long-run coefficient is negative: a one-percentage-point higher lending rate is associated with approximately 17.25% lower nominal credit in the long run. The adjustment coefficient implies slow monthly correction, while Granger tests provide no evidence of short-run predictive causality in either direction. Heteroskedasticity and RESET results require caution and further model checking.

Three policy implications follow, subject to the study's limitations. First, the Bank of Ghana should continue to balance inflation control and macroeconomic stability with the effects that borrowing costs may have on private credit. Transparent loan pricing and effective monetary transmission may help when conditions permit. Second, prudent fiscal and debt management may reduce pressures that could encourage banks to prefer government securities over private lending; however, crowding out was not directly tested. Third, banks and regulators can strengthen credit information sharing, collateral verification, loan recovery and responsible underwriting to manage lending risk and potentially improve pricing.

5.1 Limitations and Future Research

The bivariate model does not control inflation, exchange rates, fiscal conditions, monetary policy rates or bank soundness. Nominal private sector credit is a proxy for bank financing, not a direct measure of private investment, and it is not adjusted for inflation. The diagnostic evidence of heteroskedasticity and possible misspecification also limits confidence in conventional inference. Future work should verify the unusual final credit observation, use robust or otherwise appropriate inference, include additional macroeconomic controls and real-credit measures, and consider SVAR methods for identified monetary shocks, NARDL for asymmetric rate movements, and direct investment measures such as gross fixed capital formation.

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Appendix

Diagnostic figure

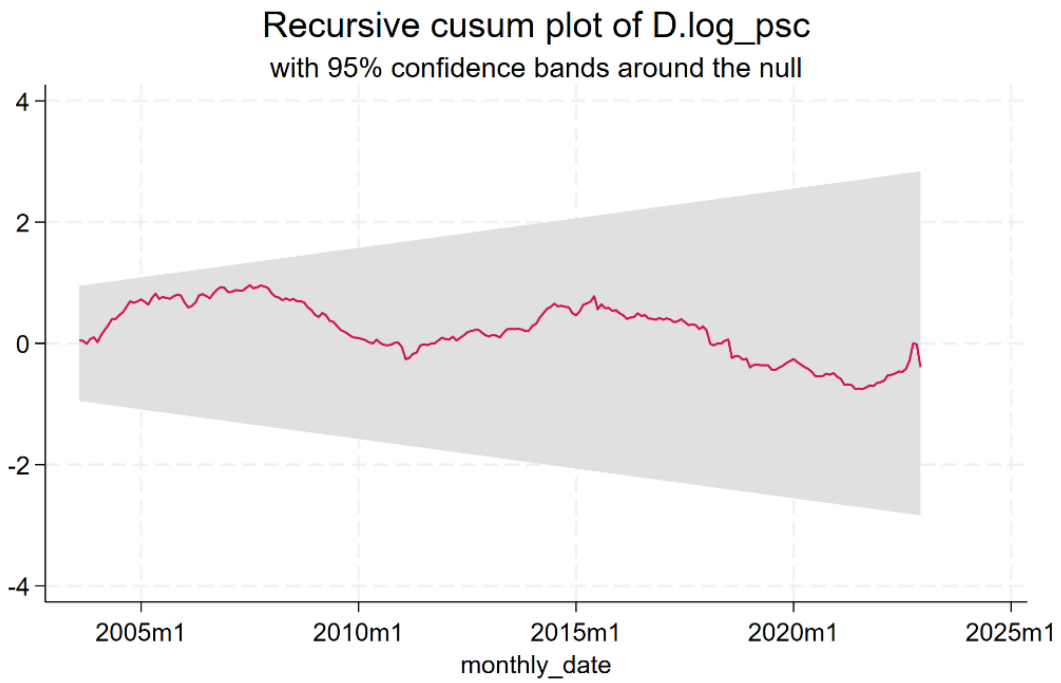


Figure 5. CUSUM stability test for the credit-growth equation (as reported in the source paper).